

A Complete Minimal Hilbert System with a Doubly Universal Element

Candidate full solution packet
arXiv:2306.11156, Questions 1 and 2

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Status. Candidate full affirmative answer to both questions, pending expert review. One and the same element for one and the same complete minimal system is universal both by its ordinary partial sums and by rearrangements. Both the source and target spaces are separable infinite-dimensional complex Hilbert spaces, and the target carries its usual norm topology.

1 Source questions

Avetisyan, Grigoryan, and Ruzhansky ask the following on printed page 8 of [1]:

which shows that $\alpha_m = c_m(f)$. $\square$

Second, the system has to be minimal in order that we can speak of unambiguously defined Fourier coefficients $c_n(f)$ of an element $f \in \mathcal{X}$. Finally, the necessity to consider various alternative kinds of universality arises mainly from the fact that for some of the most important classical systems it is known that functions universal in the usual sense do not exist (e.g., Remark 8 below). However, functions universal in various senses defined above have been shown to exist in different classical systems [8, 9, 12–17]. Nevertheless, the following questions remain open.

Question 1 *Does there exist a non-trivial triple Ω , $\mathcal{X}$ and $\{\varphi_n\}_{n=1}^\infty$ which admits a universal element $U \in \mathcal{X}$ for Ω w.r.t. $\{\varphi_n\}_{n=1}^\infty$ in the usual sense?*

Question 2 *Does there exist a non-trivial triple Ω , $\mathcal{X}$ and $\{\varphi_n\}_{n=1}^\infty$ which admits a universal element $U \in \mathcal{X}$ for Ω w.r.t. $\{\varphi_n\}_{n=1}^\infty$ in the sense of rearrangements?*

The non-triviality means that the topology of Ω is not too weak. Good choices would be $\mathcal{X} = L^1(\mathcal{M})$ and $\Omega = L^p(\mathcal{M})$, $p \in [0, 1)$, for a finite separable diffuse measure space $\mathcal{M}$, such as $\mathcal{M} = [0, 1]$ (see the discussion in Section 4). To the best of our knowledge, even for $\mathcal{M} = [0, 1]$ the above questions are open.

In their terminology, a Banach space $\mathcal{X}$ is continuously and densely embedded in a Hausdorff sequential convergence space Ω , and (φ_n) is a minimal system in $\mathcal{X}$ with biorthogonal functionals

(c_n) . An element $U \in \mathcal{X}$ is universal in the usual sense if

$$\left\{ \sum_{n=1}^N c_n(U) \varphi_n : N \in \mathbb{N} \right\}$$

has, for each $f \in \Omega$, a subsequence converging to f . It is universal in the sense of rearrangements if for every $f \in \Omega$ there is a permutation σ of $\mathbb{N}$ such that

$$\sum_{n=1}^{\infty} c_{\sigma(n)}(U) \varphi_{\sigma(n)} = f \quad \text{in } \Omega.$$

The source calls a triple non-trivial when the topology of Ω is not too weak. The construction below takes $\Omega = \ell_2$ with its ordinary Hilbert norm. More strongly, $\mathcal{X}$ is itself a Hilbert space and (φ_n) is complete as well as minimal.

2 Proof intuition

There are three moving parts. First, put in each coordinate of ℓ_2 a scaled positive and negative harmonic series, together with imaginary copies. The scalar crossing algorithm can rearrange the four subseries in coordinate j to any prescribed complex number. Their 2^{-j} scaling lets all coordinate rearrangements be interleaved without losing ℓ_2 convergence. The same labeled terms can then be put in a fixed order whose ordinary partial sums are universal.

Second, lift these target-space terms w_n through the dense diagonal map $Be_j = 2^{-j}e_j$, writing $w_n = Bv_n$. In a second Hilbert space set

$$\varphi_n = (e_n, a_n^{-1}v_n), \quad a_n = 2^{-n}.$$

The first coordinate makes the coefficient functional simply $c_n(x, y) = x_n$, so $U = (a, 0)$ has the desired Fourier terms. Completeness is the key audit: an annihilator of all φ_n would make $\sum_n |\langle v_n, z \rangle|$ finite, while the harmonic terms in any nonzero coordinate of z force it to diverge.

Finally, embed the two-coordinate Hilbert space into one copy of ℓ_2 by $I(x, y) = Ax + By$, where $A(H)$ is chosen algebraically disjoint from $B(H)$. The A -part of the Fourier series converges unconditionally, while the B -part is exactly the universal series (w_n) . It follows at once that ordinary partial sums and suitable rearrangements have the required limits.

3 A series universal in both senses

Throughout, $H = \ell_2(\mathbb{N}; \mathbb{C})$ with standard basis (e_j) . All members of the families below are treated as *labeled*; equal vectors, if any, are still distinct terms.

Lemma 1 (A rearrangement-universal Hilbert series). *For $j, m \geq 1$, put $\delta_j = 2^{-j}$ and consider the countable labeled family*

$$\mathcal{W} = \left\{ \frac{\delta_j}{m} e_j, -\frac{\delta_j}{m} e_j, \frac{i\delta_j}{m} e_j, -\frac{i\delta_j}{m} e_j : j, m \in \mathbb{N} \right\}.$$

For every $x \in H$ there is an enumeration of all terms of $\mathcal{W}$ whose series converges in H to x .

Proof. Fix $x = (x_j) \in H$. In coordinate j , order the positive and negative real terms $\pm\delta_j/m$ by the elementary crossing algorithm: take unused positive terms until the partial sum first exceeds $\operatorname{Re} x_j$, then unused negative terms until it first falls below that number, and repeat (with the roles reversed if appropriate at the start). Since both one-sided harmonic sums diverge and their terms tend to zero, every term is eventually used and the scalar partial sums converge to $\operatorname{Re} x_j$. At all times their absolute values are at most $|\operatorname{Re} x_j| + \delta_j$. Do the same with the imaginary terms and interleave the two resulting scalar orders. We obtain a local order $(u_{j,r})_{r \geq 1}$ of all four families in coordinate j such that, for

$$s_{j,R} := \sum_{r=1}^R u_{j,r},$$

we have

$$s_{j,R} \longrightarrow x_j, \quad |s_{j,R}| \leq C(|x_j| + \delta_j) \quad (1)$$

with an absolute constant C .

We now interleave the local orders. At stage k , for each $1 \leq j \leq k$ in turn, append enough as-yet-unused terms of its local order so that at least the first k local terms have been used and so that all subsequent local partial sums are within $2^{-k}\delta_j$ of x_j . This is possible by local convergence. Every labeled term is eventually used.

It remains to check convergence between, not merely at, stage endpoints. Given $\varepsilon > 0$, choose J so large that the ℓ_2 tail of $(|x_j| + \delta_j)$ beyond J is smaller than $\varepsilon/(4C + 4)$. Once the stage is sufficiently large, all local partial sums in the first J coordinates remain as close to x_j as desired. For $j > J$, whether a coordinate has not yet been started or is currently being advanced, (1) bounds its error by $(C + 1)(|x_j| + \delta_j)$. The first J errors and the uniform tail bound are therefore both small. Thus every sufficiently late global partial sum is within ε of x in H . $\square$

Lemma 2 (One order is also ordinarily universal). *The terms of $\mathcal{W}$ admit an enumeration $(w_n)_{n \geq 1}$ with both properties:*

1. *the ordinary partial sums $\sum_{n=1}^N w_n$ form a universal sequence in H ;*
2. *the series $\sum_n w_n$ is universal in H under rearrangements.*

Proof. Deleting finitely many labeled terms preserves the conclusion of Lemma 1. Indeed, if the deleted family is F and the desired sum of the remaining family is y , rearrange the full family to $y + \sum_{w \in F} w$ and delete the finitely many occurrences from that convergent series.

Fix an initial numbering of all labels and choose a sequence (d_k) in H such that every point of H is a subsequential limit; for example, repeat a countable dense set in successive finite triangular blocks. Suppose a finite ordered set of labels has been used and its sum is s . At stage k , first append the least unused label, and update s . The remaining cofinite family is rearrangement universal, so it has an enumeration whose series sums to $d_k - s$. Append a finite initial segment of that enumeration which brings the total sum within 2^{-k} of d_k .

The forced least-label step ensures that every label is eventually used, so the infinite concatenation is an enumeration (w_n) of $\mathcal{W}$. Its stage-end partial sums approach the universal target sequence (d_k) with errors tending to zero; hence the ordinary partial sums are universal. Property 2 depends only on the labeled family, not on its initial order, and follows from Lemma 1. $\square$

4 A non-trivial dense Hilbert embedding

Lemma 3. *There are bounded operators $A, B : H \rightarrow H$ such that A is an isometry, B is injective with dense range, and*

$$A(H) \cap B(H) = \{0\}.$$

Consequently

$$I : H \oplus_2 H \longrightarrow H, \quad I(x, y) = Ax + By,$$

is a bounded injective dense embedding whose inverse on its range is not continuous.

Proof. Let $Be_k = 2^{-k}e_k$. This operator is bounded and injective, and its range contains all finitely supported sequences, hence is dense.

Partition $\mathbb{N}$ into infinite disjoint sets $(S_j)_{j \geq 1}$. Choose $\gamma_j > 0$ so that

$$q_j := \gamma_j \sum_{k \in S_j} 2^{-k} e_k$$

has norm one, and define $Ae_j = q_j$. The vectors (q_j) have disjoint supports and are orthonormal, so A is an isometry.

If $Ax = By$, then for every $k \in S_j$,

$$y_k = 2^k (Ax)_k = \gamma_j x_j.$$

Because S_j is infinite, $y \in \ell_2$ forces $x_j = 0$. Thus $x = 0$ and then $y = 0$, proving the disjointness and injectivity of I . The range of I is dense because it contains $B(H)$. Finally,

$$\|(0, e_k)\|_{H \oplus_2 H} = 1, \quad \|I(0, e_k)\|_H = 2^{-k} \longrightarrow 0,$$

so the induced target topology is strictly weaker than the source norm topology. $\square$

5 The universal element

Theorem 4. *There exist separable infinite-dimensional complex Hilbert spaces $\mathcal{X}$ and Ω , a bounded injective dense embedding $I : \mathcal{X} \rightarrow \Omega$, a complete minimal system $(\varphi_n)_{n \geq 1}$ in $\mathcal{X}$ with its unique biorthogonal system $(c_n)_{n \geq 1}$, and an element $U \in \mathcal{X}$ such that its Fourier series*

$$\sum_{n=1}^{\infty} c_n(U) \varphi_n$$

is universal in Ω both in the usual sense and in the sense of rearrangements. The topology of Ω is its ordinary Hilbert norm and is strictly weaker on the embedded copy of $\mathcal{X}$ than the norm topology of $\mathcal{X}$.

Proof. Take $\Omega = H$, $\mathcal{X} = H \oplus_2 H$, and the embedding I from Lemma 3. Fix the enumeration (w_n) from Lemma 2. Every w_n is supported in one coordinate and hence belongs to $B(H)$; define

$$v_n := B^{-1}w_n.$$

As a labeled family, (v_n) consists exactly of

$$\left\{ \frac{1}{m}e_j, -\frac{1}{m}e_j, \frac{i}{m}e_j, -\frac{i}{m}e_j : j, m \in \mathbb{N} \right\}. \quad (2)$$

Let $a_n = 2^{-n}$, so $a = (a_n) \in H$, and put

$$\varphi_n = (e_n, a_n^{-1}v_n) \in \mathcal{X}, \quad c_n(x, y) = x_n, \quad U = (a, 0).$$

The c_n are continuous and $c_n(\varphi_m) = \delta_{nm}$, so (φ_n) is minimal and (c_n) is biorthogonal to it.

We verify the stronger completeness property. Suppose $(p, z) \in H \oplus_2 H$ is orthogonal to every φ_n . With the inner product taken linear in the first variable,

$$0 = \langle \varphi_n, (p, z) \rangle = \langle e_n, p \rangle + a_n^{-1} \langle v_n, z \rangle.$$

It follows from Cauchy–Schwarz that

$$\sum_{n=1}^{\infty} |\langle v_n, z \rangle| = \sum_{n=1}^{\infty} a_n |\langle e_n, p \rangle| \leq \|a\|_2 \|p\|_2 < \infty. \quad (3)$$

If $z \neq 0$, select j with $\langle e_j, z \rangle \neq 0$. The positive real subfamily in (2) then gives

$$\sum_{m=1}^{\infty} |\langle m^{-1}e_j, z \rangle| = \infty,$$

contradicting (3). Hence $z = 0$, and the orthogonality equations give $p = 0$. Thus the closed span of (φ_n) is all of $\mathcal{X}$. In particular the biorthogonal system is unique.

Since $c_n(U) = a_n$, the N th Fourier partial sum has image

$$\begin{aligned} I \left(\sum_{n=1}^N c_n(U) \varphi_n \right) &= I \left(\sum_{n=1}^N a_n e_n, \sum_{n=1}^N v_n \right) \\ &= Aa^{(N)} + \sum_{n=1}^N w_n, \end{aligned} \quad (4)$$

where $a^{(N)} = (a_1, \dots, a_N, 0, \dots)$. Given $f \in H$, ordinary universality in Lemma 2 supplies a subsequence N_k such that

$$\sum_{n=1}^{N_k} w_n \longrightarrow f - Aa.$$

As $Aa^{(N_k)} \rightarrow Aa$, (4) converges to f . This is universality in the usual sense.

For rearrangement universality, fix $f \in H$ and choose a permutation σ such that

$$\sum_{n=1}^{\infty} w_{\sigma(n)} = f - Aa$$

by Lemma 2. The orthogonal series $\sum_n a_n e_n$ converges unconditionally to a , and hence

$$\sum_{n=1}^N I(c_{\sigma(n)}(U) \varphi_{\sigma(n)}) = A \sum_{n=1}^N a_{\sigma(n)} e_{\sigma(n)} + \sum_{n=1}^N w_{\sigma(n)} \longrightarrow Aa + (f - Aa) = f.$$

This is precisely universality in the sense of rearrangements. Lemma 3 supplies all remaining embedding and non-triviality claims. $\square$

6 Scope, verification, and novelty

The theorem answers both existential questions exactly as stated, and it does so with a complete minimal system rather than a merely minimal one. It does *not* resolve the more concrete problem suggested by the authors, namely whether one may take $\mathcal{X} = L^1(\mathcal{M})$ and $\Omega = L^p(\mathcal{M})$ for $0 \leq p < 1$, nor does it produce a classical trigonometric or Walsh system. The construction is deliberately tailored.

The proof was audited at the following pressure points: the countable interleaving converges at every partial sum rather than only at block endpoints; a cofinite deletion really preserves rearrangement universality; $A(H) \cap B(H) = \{0\}$; the annihilator argument gives completeness; and the same permutation is compatible with the unconditionally convergent first-coordinate series. The accompanying `verification.md` records these checks. No numerical step is used as proof.

A bounded novelty search was performed on 11 August 2026. The run registry, solution, attempt, and proof-gap indexes were searched for arXiv:2306.11156, the exact title, and the core phrases “universal element”, “minimal system”, and “universal by rearrangements”. The locally parsed arXiv corpus and a current arXiv-domain web search were searched for the exact question and close phrases. These searches recovered the source paper and background work on universal *series*, but no later paper answering the element questions. This is not an exhaustive bibliographic review, so novelty remains subject to specialist confirmation.

Human-review recommendation. Review as a candidate full solution. The highest-value checks are Lemma 1’s uniform tail control, Lemma 2’s cofinite scheduling, and the intended scope of the source phrase “non-trivial triple.”

References

- [1] Z. Avetisyan, M. Grigoryan, and M. Ruzhansky, *Elements (functions) that are universal with respect to a minimal system*, arXiv:2306.11156 (2023), Questions 1 and 2 on printed page 8.